CONVOLUTION, CHARACTERISTIC FUNCTION, FOURIER TRANSFORM

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Contents

1. Convolution Interpretation

I follow [Fol92, p 207]

From the definition

$$(f * g)(x) = \int f(x-y)\, g(y)\, dy = \int f(y)\, g(x-y)\, dy, \tag{1}$$

one has two interpretations.

In both cases think of f as the "main" function and g as the subsidiary "weighting" or "mollifying" function.

1.1. Linear Combination of Translates.

From the first integral in (1), using Riemann sums:

$$(f * g)(x) = \int f(x-y)\, g(y)\, dy \approx \sum_i f(x-y_i)\, g(y_i)\Delta y_i.$$

So

(2) *The function $f * g$ is the continuous superposition (linear combination) of translates of f by y_i with coefficients $g(y_i)\Delta y_i$.*

1.2. Weighted Average.

From the second integral in (1),

$$(f * g)(x) = \int f(y)\, g(x-y)\, dy.$$

Consider the case $g \geq 0$ and $\int g = 1$. Then

(3) *The value $(f * g)(x)$ is the weighted average of f using the weight function $w(\cdot) = g(x - \cdot)$.*

The weight function $g(x - \cdot)$ is obtained from g by reflecting the graph of g in the vertical axis and then translating by x. See Figure 1.

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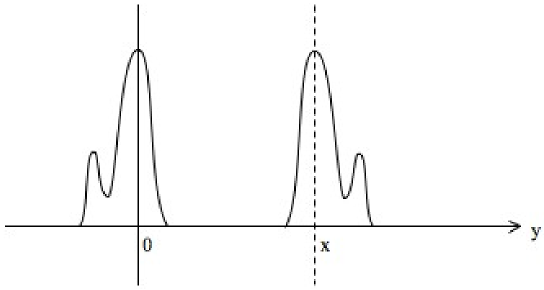

FIGURE 1. The graph of $y \mapsto g(y)$ is on the left and the graph of $y \mapsto g(x-y)$ is on the right.

For example, if

$$g = \frac{1}{2a}\mathcal{X}_{[-a,a]}, \quad g(x-\cdot) = \frac{1}{2a}\mathcal{X}_{[x-a,x+a]}(\cdot),$$
$$\text{then} \quad (f*g)(x) = \frac{1}{2a}\int_{x-a}^{x+a} f.$$

2. Fourier transform of nice function

I follow [CH53, pp 77–79].

2.1. Motivation for inversion formula. Suppose $f \in L^1(\mathbb{R})$, continuous and piecewise C^1.

The Fourier series on $[-L, L]$ is

$$f(x) = \sum_{n=-\infty}^{n=\infty} a_n e^{in\frac{\pi}{L}x},$$
$$a_n = \frac{1}{2L}\int_{-L}^{L} f(t)e^{-in\frac{\pi}{L}t}\,dt.$$

(Convergence is both pointwise and in L^2 sense.)

Hence on $[-L, L]$,

$$f(x) = \frac{1}{2\pi}\sum_{n=-\infty}^{n=\infty} \frac{\pi}{L}\int_{-L}^{L} f(t)e^{in\frac{\pi}{L}(x-t)}\,dt.$$

Let $L \to \infty$ and thinking of π/L as δu it is *plausible* that

$$\begin{aligned} f(x) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} f(t)e^{iu(x-t)}\,dt\right) du \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} f(t)e^{-iut}\,dt\right) e^{iux}\,du. \end{aligned} \tag{4}$$

The *Fourier transform* is defined by

$$\widehat{f}(u) := \int_{-\infty}^{\infty} f(t)e^{-iut}\,dt,$$

and so we can write (4) as the inversion formula

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(u)e^{iux}\,du. \tag{5}$$

2.2. Proof of inversion formula. Instead of rigorously justifying the limit used to obtain (4) it is easier to prove (4) directly as follows.

Proof.

$$f(x) = \lim_{v\to\infty}\frac{1}{\pi}\int_{-a}^{a} f(x-t)\frac{\sin vt}{t}\,dt \tag{6}$$

($\frac{\sin vt}{t}$ is essentially the Dirichlet kernel – give reference ***)

$$\begin{aligned}
&= \lim_{v\to\infty} \frac{1}{\pi} \int_{-a}^{a} f(x-t) \left(\int_0^v \cos ut \, du \right) dt \\
&= \lim_{v\to\infty} \frac{1}{\pi} \int_0^v \left(\int_{-a}^{a} f(x-t) \cos ut \, dt \right) du \\
&= \frac{1}{\pi} \int_0^\infty \left(\int_{-a}^{a} f(x-t) \cos ut \, dt \right) du \\
&\to \frac{1}{\pi} \int_0^\infty \left(\int_{-\infty}^{\infty} f(x-t) \cos ut \, dt \right) du \quad \text{as } a \to \infty
\end{aligned}$$

(since using (6) the error is bounded by $\frac{1}{2\pi a}\|f\|_{L^1}$)

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(x-t) e^{iut} \, dt \right) du$$

(since $\sin ut$ is odd in u)

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(t) e^{iu(x-t)} \, dt \right) du$$

This is (4). □

3. Characteristic Function

I follow [Ash00, pp 291–298]

3.1. **Definitions.** Suppose μ is a probability measure (more generally, a finite measure) on $\mathcal{B}(\mathbb{R})$.

The *probability distribution* F is defined either as a measure or as a function by

$$\mu(a,b] = F(a,b] = F(b) - F(a).$$

So F is increasing, continuous from the right, and has a limit from the left.

The *characteristic function* h is

$$h(u) = \int e^{iux} \, d\mu(x) = \int e^{iux} \, dF(x) = \mathbb{E}e^{iuX}, \tag{7}$$

where in the last term, X is any r.v. with dist $X = F$.

If F is C^1 this is just the Fourier transform of the corresponding probability density dF/dx with a sign change.

3.2. **Inversion formula.**

Proposition 3.1. *Suppose h is the characteristic function of the distribution F, and suppose $h \in L^1$. Then*

$$f(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} h(u) \, du \tag{8}$$

is a density for F.

One always *has (without assuming $f \in L^1$)*

$$F(a,b] = \lim_{c\to\infty} \frac{1}{2\pi} \int_{-c}^{c} \frac{e^{-iua} - e^{-iub}}{iu} h(u) \, du, \tag{9}$$

if F is continuous at a and b.

Remark: Formally, (9) comes from (8) by integrating. But it holds more generally.

Proof. Let

$$
\begin{aligned}
I_c &= \frac{1}{2\pi}\int_{-c}^{c} \frac{e^{-iua}-e^{-iub}}{iu} h(u)\,du \\
&= \frac{1}{2\pi}\int_{-c}^{c} \frac{e^{-iua}-e^{-iub}}{iu} \left(\int_{-\infty}^{\infty} e^{iux}\,dF(x)\right) du \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} \left(\int_{-c}^{c} \frac{e^{-iua}-e^{-iub}}{iu} e^{iux}\,du\right) dF(x) \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} \left(\int_{-c}^{c} \frac{\sin u(x-a)-\sin u(x-b)}{u}\,du\right) dF(x)
\end{aligned}
$$

(the "cos part" is odd and so integrates to 0)

$$
= \frac{1}{2\pi}\int_{-\infty}^{\infty} \left(\int_{-c(x-a)}^{c(x-a)} \frac{\sin v}{v}\,dv - \int_{-c(x-b)}^{c(x-b)} \frac{\sin v}{v}\,dv\right) dF(x).
$$

Since $\int_{-\infty}^{\infty} \frac{\sin v}{v}\,dv = \pi$, as $c \to \infty$, the inner term converges to $2\pi\mathcal{X}_{(a,b]} + \pi\mathcal{X}_{\{a\}} - \pi\mathcal{X}_{\{b\}}$, . By the D.C.T.

$$
I_c \to F(a,b] + \frac{1}{2}f\{a\} - \frac{1}{2}f\{b\},
$$

proving a bit more than stated in (9).

Next assume h is integrable.
Then f *defined* by (8) is *continuous* by the D.C.T.
Moreover

$$
\begin{aligned}
\int_a^b f(x)\,dx &= \frac{1}{2\pi}\int_{-\infty}^{\infty} h(u)\left(\int_a^b e^{-iux}dx\right) du \\
&= \lim_{c\to\infty}\int_{-c}^{c} h(u)\frac{e^{-iua}-e^{-iub}}{iu}\,du \\
&= F(a,b], \qquad (10)
\end{aligned}
$$

provided a and b are continuity points of F , i.e. $F\{a\} = F\{b\} = 0$.

Since the discontinuities of F are countable, the continuous points are dense. It follows $F\{c\} = 0\ \forall c$, and so (10) is true for all a and b.

It follows F is differentiable everywhere and $F' = f$. □

3.3. **Properties of the characteristic function.**

(1) Suppose $S = X_1 + \cdots X_n$ where the X_i are independent with characteristic function h_i. Then S has characteristic function $h = h_1 \times \cdots \times h_n$.
Proof: $\mathbb{E}e^{iu(X_1+X_2)} = \mathbb{E}e^{iuX_1}\mathbb{E}e^{iuX_2}$ by independence.

(2) $|h(u)| \le h(0) = 1$.
Proof: Immediate

(3) h is continuous.
Proof: By D.C.T.

(4) $h(-u) = \overline{h(u)}$.
Proof: Immediate.

(5) h is real valued if F is symmetric. (In fact, "iff")
Proof: Since $\overline{h(u)} = \int e^{-iux}\,dF(x) = \int e^{iux}\,dF(x) = h(u)$, where the second equality is by the symmetry.

(6) If $\mathbb{E}|X|^k < \infty$ then $h \in C^1$, $h^{(k)}(u) = \int (ix)^k e^{iux}\,dF(x)$, $\mathrm{h}^k(0) = i^k\mathbb{E}X^k$.
Proof: The $k = 1$ case follows from the D.C.T. after applying the mean value theorem. Then use induction for $k \ge 1$.

(7) *Bochner's Theorem*: h is a characteristic function iff h is continuous at 0 and *nonnegative definite*.
That is, $\sum_{j,k=1}^{n} h(u_j - u_k)a_j\overline{a}_k$ is real and ≥ 0 for all n, all $u_1, \ldots, u_n \in \mathbb{R}$, and all $a_1, \ldots, a_n \in \mathbb{C}$.

Proof: If h is a characteristic function it is continuous everywhere by (3). Also

$$0 \leq \int \left| \sum_{j=1}^{n} a_j e^{iu_j x} \right|^2 dF(x) = \sum_{j,k=1}^{n} \int a_j e^{iu_j x} \overline{a}_k e^{-iu_k x} dF(x) = \sum_{j,k=1}^{n} h(u_j - u_k) a_j \overline{a}_k.$$

The other direction is non trivial.

References

[Ash00] Robert B. Ash, *Probability and measure theory*, 2nd ed., Harcourt/Academic Press, Burlington, MA, 2000. With contributions by Catherine Doléans-Dade.

[CH53] R. Courant and D. Hilbert, *Methods of mathematical physics. Vol. I*, Interscience Publishers, Inc., New York, N.Y., 1953.

[Fol92] Gerald B. Folland, *Fourier analysis and its applications*, Wadsworth & Brooks/Cole, 1992.